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System and method for blocking a luminal body

Abstract

A surgical system for blocking a luminal body at a selected location has a tubular body that extends from a proximal end to a distal end, and a blocking device on the distal end of the tubular body. The blocking device has a guide cap and a gripping feature, and at least one annular inflatable portion positioned around an outer circumference of the gripping feature. A first conduit through the tubular body is connected to a pressure source for inflating the annular inflatable portion, and a second conduit through the tubular body is connected to a vacuum source for suctioning a luminal wall of the luminal body against the gripping feature.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application for a utility patent is a continuation of a previously filed utility patent, now pending, having the application Ser. No.

18/981,354, filed Dec. 13, 2024, which claims the benefit of U.S. Provisional Application No. 63/609,856, filed Dec. 13, 2023.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) This invention relates generally to surgical systems and methods, and more particularly to a surgical system that includes a blocking device for blocking a luminal body such as a blood vessel or similar tubular body structure.

Description of Related Art

(2) There is a need in the medical field to block a luminal body, such as a blood vessel or similar tubular body structure, in this embodiment for blocking blood flow during a surgery. In one embodiment, the blocking device may be used to block the aorta of a person or animal, for treating an aortic aneurysm (AA), although this example is illustrative and not limiting, and the invention may be used in a similar fashion for a wide variety of uses.

(3) An AA is characterized by permanent full-thickness dilation of the aortic wall, greater than 50% in diameter of normal size, and it can be generally classified into thoracic aortic aneurysm (TAA), abdominal aortic aneurysm (AAA), or other forms, according to the involved segments. Although most AAs are asymptomatic at the time of diagnosis, the incidence of complications increases as the aneurysm expands. The complications of AA, dissection and rupture, are usually catastrophic, with an almost 100% mortality rate. For elective AAA repair, the 28-day mortality rate was reported to be 3.3%-27.1% in men and 3.8%-54.3% in women in a Dutch population. In China, the overall 30-day mortality of infrarenal AAA repair was approximately 8.8% in a single vascular center.

(4) In an investigation of the general American population, the AAA-related mortality rate was 2.2 deaths per 100,000 in 2016. AA is one of the major cardiovascular diseases with an increased number of years of life lost (YLLs) and deaths globally.

(5) Treatment is typically accomplished via open heart surgery wherein the chest is opened, the heart is stopped, and then the damaged part of the aorta is replaced with a graft. This process is obviously extremely difficult to perform given the time constraints in stopping the heart, as well as traumatic to the patient, and often leads to fatal outcomes.

(6) There is a need in the art for an alternative method of treatment that does not involve stopping the heart of the patient to make this repair. The present invention fulfills these needs and provides further advantages as described in the following summary.

SUMMARY OF THE INVENTION

(7) The present invention teaches certain benefits in construction and use which give rise to the objectives described below.

(8) The present invention provides a surgical system for blocking a luminal body at a selected location. The surgical system comprises a tubular body that extends from a proximal end to a distal end, and a blocking device on the distal end of the tubular body. The blocking device has a guide cap and a gripping feature, and at least one annular inflatable portion positioned around an outer circumference of the gripping feature. A first conduit through the tubular body is connected to a pressure source for inflating the annular inflatable portion, and a second conduit through the tubular body is connected to a vacuum source for suctioning a luminal wall of the luminal body against the gripping feature. Additional conduits may be provided for other purposes, such as allowing fluid repositioning, as discussed below.

(9) A primary objective of the present invention is to provide a surgical system having advantages not taught by the prior art.

(10) Another objective is to provide a surgical system that allows an aorta to be blocked for a repair procedure without the need to stop the heart of the patient.

(11) A further objective is to provide a surgical system having a blocking device that can be

inserted through an incision and then inflated to grip a luminal wall and block a luminal body.
(12) Other features and advantages of the present invention will become apparent from the following more detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings illustrate the present invention.
(2) FIG. **1** illustrates a blocking device for blocking a luminal body, in this embodiment part of a surgical system for repairing an aortic aneurysm according to one embodiment of the present invention, illustrating the blocking device being positioned in an aorta of a heart upstream from an aneurism, to block fluid flow through the aorta.
(3) FIG. **2** is an up-close view of the blocking device inserted into the aorta so that gripping features are positioned within the aorta in a deflated state.
(4) FIG. **3** is an up-close view of the blocking device once the gripping features have been inflated to an inflated state.
(5) FIG. **4** is a side elevation cross-sectional view of a second embodiment of the blocking device, illustrating the gripping features in a partially inflated state.
(6) FIG. **5** is a cross-sectional view taken along lines **5-5** in FIG. **4**, illustrating the gripping feature in an inflated state.
(7) FIG. **6** is a cross-sectional view taken along lines **6-6** in FIG. **4**, illustrating dividing walls circumferentially spaced to divide an inflatable portion of the gripping feature into a plurality of chambers.
(8) FIG. **7** is a side cross section of another embodiment of a gripping feature, illustrating the gripping feature in a deflated state.
(9) FIG. **8** is a side cross section of the gripping feature of FIG. **7**, illustrating the gripping feature in an inflated state.
(10) FIG. **9** is a side cross section of the gripping feature of FIG. **4**, illustrating a fluid pump system integrated into the device.
DETAILED DESCRIPTION OF THE INVENTION
(11) The above-described drawing figures illustrate the invention, a blocking device used to perform a surgical procedure within a luminal body of a person or other animal. In the present embodiment, the device is illustrated being used for repairing an aortic aneurysm in a human; however, those skilled in the art may use this device in any luminal body for a wide range of surgical procedures.
(12) FIG. **1** illustrates one embodiment of a blocking device **20** for blocking a luminal body **14**, in this embodiment part of a surgical system **10** for repairing an aortic aneurysm **16** according to one embodiment of the present invention. FIG. **1** illustrates the blocking device **20** being positioned in an aorta of a heart **12** upstream from an aneurism, to block fluid flow through the aorta. The blocking device **20** may be used in any artery or vein, or any other luminal body.
(13) The system of FIG. **1** includes the blocking device **20** shown inside of the luminal body **14** that has the aneurysm **16**. The blocking device **20** is adapted to be inserted into the luminal body **14** through an incision (not shown) to treat the aneurysm **16** by temporarily blocking the aorta so that the aneurysm **16** may be repaired, without requiring the heart **12** to be stopped. Since the heart remains beating and supplying blood to the brain, the surgery may last up to 20 minutes, making the process much easier for the surgeon, and less traumatic to the patient, resulting in much better outcomes.
(14) The repair procedure may involve a resecting of the tissue, the placement of a stent (not

shown) or other suitable device inside the aneurysm **16**, or any other similar or equivalent surgical procedure known in the art. While one particular embodiment is discussed, in other embodiments, the blocking device **20** may be used to operate on other luminal bodies, such as for other vascular problems, tumor removal, embolisms, and the like, in arteries, veins, abdominal cavities, or other part of the body that could benefit from the present invention as-described, and any similar or equivalent procedures should be considered within the scope of the present invention.
(15) FIG. **2** is an up-close view of the blocking device **20** inserted into the aorta so that gripping features **40** are positioned within the aorta in a deflated state. FIG. **3** is an up-close view of the blocking device **20** once the gripping feature **40** has been inflated to an inflated state. As shown in FIGS. **1-3**, the surgical system comprises a tubular body **22** that extends from a proximal end **24** to a distal end **26**, and the blocking device **20** on the distal end **26** of the tubular body **22**. The tubular body **22** is sized and shaped to be inserted into the aorta through an incision(s), and fed through a blood vessel to the aneurysm **16**, where the blocking device **20** may be positioned past and adjacent the aneurysm **16** to block the aorta for repair. For purposes of this application, the term "adjacent" is defined to mean a suitable distance for performing the operation described below, as determined by the physician. Several embodiments of the gripping feature **40** are discussed in the following Figure descriptions. The blocking device **20** has a guide cap **28** at the distal end **26**, and the gripping feature **40** has at least one annular inflatable portion **44** around an outer circumference of the gripping feature **40**. In this embodiment, the guide cap **28** has a conical or an arcuate shape, but any suitable shape or structure known in the art may be used.
(16) The gripping feature **40** is adapted for establishing a firm gripping connection with an inner surface **18** of the luminal body **14**, for anchoring the device in place. In this embodiment, the gripping feature includes at least one annular plug **42** that extends radially from the tubular body **22** and terminates in the inflatable portion **44**. The annular plugs **42** and annular inflatable portions **44** are concentrically aligned around the axis of the tubular body **22**, best shown in FIG. **6**. In the embodiment of FIGS. **2-3**, the annular inflatable portions **44** each have a symmetrical tubular shape, but other shapes and structures may be constructed, other examples being shown in FIGS. **4-9** and discussed below. While three inflatable portions **44** are illustrated, any suitable number may be included, including one, which should be considered within the scope of the present invention.
(17) As illustrated in FIG. **2**, a first conduit **30** extends through the tubular body **22** and is connected to a pressure source **32** for inflating the annular inflatable portion **44**. In this embodiment, the first conduit **30** passes up the tubular body **22** and through the annular plugs **42** to access the annular inflatable portions **44**. As shown in FIG. **3**, a second conduit **34** extends through the tubular body **22** and is connected to a vacuum source **36**, i.e., a source of lower pressure which is lower than atmospheric pressure, for suctioning a luminal wall of the luminal body **14** against the gripping feature **40**, discussed in greater detail below. In this embodiment, the tubular body **22** includes a plurality of valves **38** positioned between each annular plug **42** so that the inner surface **18** of the luminal body **14** may be suctioned to the blocking device **20**, thereby effectively blocking the aneurysm **16** from, for example, the passage of fluids from upstream. In some embodiments, the valves **38** may be closeable, so that once suctioned, the vacuum may be shut off without losing the achieved blocking function. Furthermore, there may be some form of closable valve (not shown) of the first conduit **30**, so that the pressure source **32** may cease its input without losing inflation of the inflatable portions **44**. In further embodiments, the pressure source **32** and the vacuum source **36** may be separate structures, or they may be integrated into a single structure/device.
(18) FIG. **4** is a side elevation cross-sectional view of a second embodiment of the blocking device **20**, illustrating the gripping feature **40** in a deflated state, and FIG. **5** is a cross-sectional view taken along lines **5-5** in FIG. **4**, illustrating the gripping feature **40** in an inflated state. As shown in FIGS. **4-5**, in this embodiment, the inflatable portions **44** each include an inner layer **48** forming an inner body **50**, and an outer layer **52** forming an outer body **54**, which may be recessed in a concave

shape in the deflated state, and expanded so that the outer layer **52** contacts the inner surface **18** of the luminal body **14** in an inflated state.

(19) Each outer layer **52** may include a clamping area **56** around the outer periphery that includes a "tread" **58**, in this embodiment being in the form of a plurality of posts **58**, wherein a plurality of holes **60** may be formed between the posts **58** on the clamping area **56**. While posts are illustrated and described, any form of tread may be constructed, e.g., ridges, waves, cones, etc. The tread **58** may be formed of a pliable material (e.g., rubber, etc.). Furthermore, each inner layer **48** may include a plurality of spacers **62** around the entire outside of the inner layer **48**, in this embodiment also being in the form of a plurality of posts.

(20) As illustrated, in this embodiment, the second conduit **34** extends into the inflatable portion(s) **44**, rather than between the inflatable portions **44**, for suctioning the clamping area **56** against the inner surface **18** of the luminal body **14**. The first conduit **30** communicates with the inner body **50**, and the second conduit **34** communicates with the outer body **54**, outside of the inner body **50**. In this manner, pressure may be applied to the inner body **50** to inflate it, which presses the clamping area **56** of the outer layer **52** against the luminal body **14**. The plurality of spacers **62** maintain a separation between the inner and outer layers **48** and **52**, so that vacuum may then be introduced between the inner and outer layers **48** and **52** to suction the clamping area **56** against the luminal body **14**. The inner layer **48** does not include holes, allowing air to pass between the inner and outer layers **48** and **52**, while maintaining pressure of the inner body **50**. As shown in FIG. **5**, in some embodiments, side ridges **64** may be formed on outer edges of the clamping area **56** to seal the area between the clamping area **56** and the luminal body **14** to contain the vacuum force. In this embodiment, the holes **60** are formed in a "zig-zag" configuration across the clamping area **56**, but the holes **60** could potentially be formed in any suitable configuration, as determined by one skilled in the art.

(21) FIG. **6** is a cross-sectional view of the blocking device **20** taken along lines **6-6** in FIG. **4**, illustrating dividing walls **66** circumferentially spaced to divide each of the inflatable portions **44** into a plurality of chambers **68**. As shown in FIG. **6**, the dividing walls **66** may extend radially from the tubular body **22** of the blocking device **20**, dividing the annular plug **42** and the annular inflatable portion **44** into quadrants. While quadrants are illustrated, any suitable number of chambers **68** may be implemented for the purpose described below. In this embodiment, the first and second conduits **30** and **34** are also split into the same number of valves as there are chambers **68**, for inflating and vacuuming each chamber **68** as needed. In FIG. **6**, the holes **60** for suctioning are not visible, but this is only for the purpose of clarity.

(22) In a typical use where the blocking device **20** inserted, there may be excess fluid such as blood seeping/flowing from a portion of the inner surface **18** of the luminal body **14**, i.e., if there is a lesion in the inner surface **18**, or similar. To prevent the vacuum continuously aspirating the fluid into the entire inflatable portion **44** and thereby preventing proper clamping of the clamping surface, only one of the chambers **68** may instead be prevented from clamping, wherein the remaining chambers **68** can clamp sufficiently to allow the luminal body **14** to be blocked for a surgical procedure. In some cases, the valve of the second conduit **34** (connected to the vacuum source **36**) may be closed off in the affected chamber, so fluid is not drawn from that chamber. In cases where there is a plurality of tiered inflatable portions, a single chamber of one of the inflatable portions **44** failing to suction will not pose an issue for the procedure.

(23) FIG. **7** is a side cross section of another embodiment of the gripping feature **70**, illustrating the gripping feature **70** in a deflated state, and FIG. **8** is a side cross section of the gripping feature of FIG. **7**, illustrating the gripping feature **70** in an inflated state. As shown in FIGS. **7-8**, in this embodiment, the gripping feature **70** may include an inner body **72** that receives vacuum from the vacuum source **36** and which includes holes **74** formed in the clamping surface, while an outer body **76** receives pressure from the pressure source **32**. In this embodiment, the spacers **62** of FIGS. **4-5** are not necessary for separating the inner and outer bodies to allow vacuum to pass

therebetween. While FIG. **8** illustrates that the clamping surface may remain semi-concave in the inflated state, the arrangement of the inner and outer bodies **72** and **76** does not necessitate this, and it is intended only for the purpose of example. In this case, the side ridges **64** formed on outer edges of the clamping area **56** may also seal the area between the clamping area **56** and the luminal body **14** to contain the vacuum force.

(24) FIG. **9** is a side cross section of the gripping feature **40** of FIG. **4**, illustrating a fluid pump system **78** integrated into the device. As shown in FIG. **9**, in this embodiment, the fluid pump system **78** may include an intake tube **80** (or port, hose, or other intake mechanism) that draws fluid from beneath the blocking device **20** and pumps it into a pump conduit **82** that extends from the proximal end **24** of the tubular body **22** and through to the distal end **26**, to an outlet **84** of the guide cap **28**. The fluid pump system **78** may be a hand pump, manual stroke, syringe, electric pump, etc., that pushes blood downstream to upstream. The pump **78** could be inside of the blocking device **20**, or outside, depending on the needs of the medical professional.

(25) If used with blood, the fluid pump system **78** functions to draw spilled blood present in the lower aorta or surrounding area, and pumps it into the upper aorta from where it can provide potentially vital support to the patient, who may be facing serious injury or death.

(26) The title of the present application, and the claims presented, do not limit what may be claimed in the future, based upon and supported by the present application. Furthermore, any features shown in any of the drawings may be combined with any features from any other drawings to form an invention which may be claimed.

(27) As used in this application, the words “a,” “an,” and “one” are defined to include one or more of the referenced item unless specifically stated otherwise. The terms “approximately” and “about” are defined to mean+/−10%, unless otherwise stated. Also, the terms “have,” “include,” “contain,” and similar terms are defined to mean “comprising” unless specifically stated otherwise. Furthermore, the terminology used in the specification provided above is hereby defined to include similar and/or equivalent terms, and/or alternative embodiments that would be considered obvious to one skilled in the art given the teachings of the present patent application. While the invention has been described with reference to at least one particular embodiment, it is to be clearly understood that the invention is not limited to these embodiments, but rather the scope of the invention is defined by claims made to the invention.

Claims

1. A surgical system for blocking a luminal body at a selected location, the surgical system comprising: a tubular body that extends from a proximal end to a distal end; a blocking device on the distal end of the tubular body, the blocking device having a guide cap and a gripping feature; at least one annular inflatable portion around an outer circumference of the gripping feature; a first conduit through the tubular body connected to a pressure source for inflating the annular inflatable portion; and further comprising a second conduit through the tubular body connected to a vacuum source for suctioning the luminal wall of the luminal body against the gripping feature.
2. The surgical system of claim 1, wherein the gripping feature includes at least one annular plug that extends radially from the tubular body and terminates in the inflatable portion.
3. The surgical system of claim 1, wherein the at least one inflatable portion includes an inner layer forming an inner body, and an outer layer having at least one hole and forming an outer body, and wherein the first conduit communicates with the inner body, and the second conduit communicates with the outer body, so that pressure may be applied to inflate the inner body, which presses the inflatable portion against the luminal body, and vacuum may be applied to the outer body to suction the inflatable portion to the luminal body.
4. The surgical system of claim 3, wherein a plurality of spacers on the outside of the inner layer maintains a separation between the inner and outer layers.

5. The surgical system of claim 1, further comprising a plurality of dividing walls extending radially from the tubular body within the inflatable portion, the dividing walls being circumferentially spaced to form a plurality of chambers in the inflatable portion.
6. The surgical system of claim 1, further comprising a pump having an intake tube that draws fluid from downstream of the blocking device and pumps it through a pump conduit and out of an outlet upstream of the blocking device.
7. A method for blocking a luminal body at a selected location, the method comprising the steps of: providing a blocking device having a tubular body that extends from a proximal end to a distal end, the distal end having a guide cap and at least one gripping feature, wherein each of the at least one gripping feature has an annular inflatable portion around an outer circumference of the gripping feature, and further comprising a second conduit through the tubular body to the inflatable portion; inserting the blocking device into the luminal body, and guiding it through the luminal body to the selected location; inflating the at least one annular inflatable portion of the gripping feature from a deflated state that is smaller than the diameter of the luminal body, to an inflated state that is large enough to frictionally engage the luminal body, thereby blocking the luminal body; and applying vacuum through the second conduit for suctioning the luminal wall of the luminal body against the gripping feature.
8. The method of claim 7, wherein the luminal body is an artery or vein.
9. The method of claim 7, further comprising the step of performing a surgical procedure on or near the luminal body.